Mohamed Amara

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Education

CUNY College of Staten Island Expected May 2027
B.S. in Computer Science *Staten Island, NY*
- **Relevant Coursework:** Data Structures & Algorithms, Network Security, Operating Systems, Digital Logic, Discrete Mathematics

Experience

Software Engineering Intern Mar 2025 – May 2025
Auristor *Washington, DC*
- Built a Go monitoring system from scratch that collected 150+ infrastructure metrics (CPU, memory, disk I/O, network throughput) into Grafana dashboards, cutting average issue detection time by roughly 40%.
- Turned Figma mockups into production React components with Tailwind CSS, working through several review cycles with the design team on spacing, typography, and responsive behavior.
- Set up Docker containers and Jenkins CI/CD for the monitoring stack, giving the team weekly releases with zero-downtime deployments.
- Trained 8 teammates on reading the dashboards and responding to alerts, and wrote runbooks for common failure scenarios.

Cybersecurity Intern Feb 2023 – May 2023
CodePath *New York, NY*
- Hardened learning platform servers with input sanitization and parameterized queries, blocking SQL injection and XSS attempts across all user-facing endpoints.
- Ran penetration tests on 10+ web applications with Burp Suite, OWASP ZAP, and Nmap, finding everything from insecure authentication to misconfigured CORS policies.
- Wrote up findings with severity ratings and reproduction steps, then worked with developers to verify the patches.

Projects

Memories | *Go, TypeScript, PostgreSQL, FFmpeg, AWS* | memories.brozy.org | Shaku-Med/free_file Ongoing
- Building a video platform solo: a React Router 7 and TypeScript web app for watch pages, reels, and a creator studio, backed by four services that handle uploading, transcoding, moderation, and search.
- Wrote the Go upload service that turns raw uploads into HLS renditions and thumbnails with FFmpeg, plus a playback service that signs segment tokens so video files cannot be hotlinked.
- Built the feed, reels, and recommendations as PostgreSQL stored functions, with a Python embedding service that vectorizes titles and descriptions so search can surface similar videos.
- Added NSFW scoring on upload, dual storage across Cloudflare R2 and GitHub with a weekly quota ledger, and deployment on EC2 behind nginx with Redis job queues and GitHub Actions CI/CD.

ASCILINE | *Python, FastAPI, WebSockets, NumPy, OpenCV* | YusufB5/ASCILINE Ongoing
- Contributor to an open source real-time ASCII video streaming engine (2.5k stars, 280+ forks) that renders video server side and streams binary text frames over WebSocket to an HTML5 canvas player at 30 FPS.
- Working on the streaming and rendering path, where frames are compressed per frame with delta and run-length encodings and decoded client side without any browser video codec involved.

HLS Grabber | *JavaScript, Chrome Extensions, Python, FFmpeg* | Shaku-Med/hls-downloader 2026
- Built a Chrome extension that inspects network traffic per tab, filters the noise down to real media manifests, and saves a stream in one click instead of stitching segments together by hand.
- Wrote the Python native messaging host that runs FFmpeg to write files to disk, an installer that registers it on Windows and Linux, and a job queue so downloads keep running after the popup closes.

Technical Skills

Languages: Python, Go, JavaScript, TypeScript, Java, Kotlin, C++, SQL, PHP

Frontend: React, Next.js, React Router 7, Tailwind CSS, Jetpack Compose, HTML5, CSS3

Backend & Infra: FastAPI, Flask, Node.js, Express.js, Docker, Jenkins, nginx, Redis, AWS (EC2, S3), Cloudflare R2

Tools & Frameworks: Git, GitHub Actions, Grafana, WebRTC, FFmpeg, OpenCV, TensorFlow, Supabase, PostgreSQL, Burp Suite